Y1 Chapter 11 Vectors

1: Add/scale factors and show vectors are parallel.

2: Calculate magnitude and direction of a vector.

3: Understand and use position vectors.

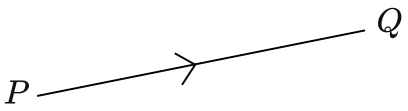
4: Solve both geometric problems.

5: Understand speed vs velocity

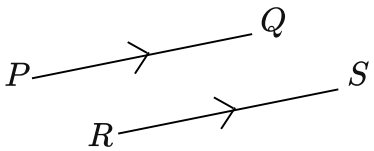
A Whereas a **coordinate** represents a **position** in space, a **vector** represents a **displacement** in space.

A vector has 2 properties:
- Direction
- Magnitude (i.e. length)

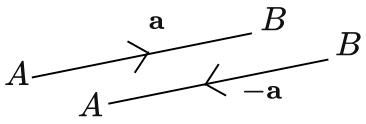
If P and Q are points then $\vec{PQ}$ is the vector between them.



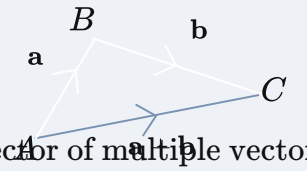
B If two vectors $\vec{PQ}$ and $\vec{RS}$ have the same magnitude and direction, **they're the same vector** and are **parallel**.



C $\vec{AB} = -\vec{BA}$ and the two vectors are parallel, equal in magnitude but in **opposite directions**.



D Triangle Law for vector addition: $\vec{AB} + \vec{BC} = \vec{AC}$



The vector of multiple vectors is known as the **resultant vector.**
(you will encounter this term in Mechanics)

Have you ever wondered what happens if you 'multiply' two vectors or two sets?

In KS2/3 you probably only experienced variables holding numerical values. You since saw that variables can represent other mathematical types, such as sets or vectors:

$$x = 3$$
$$\mathbf{a} = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$$
$$T = \{2, 5, 9, 10\}$$

But when we do we need to define **explicitly** what operators like '+' and '×' mean.

$$1 + 3 = 4 \qquad \begin{pmatrix} 3 \\ -2 \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 4 \\ -2 \end{pmatrix}$$
$$\begin{pmatrix} 3 \\ 2 \\ -1 \end{pmatrix} \times \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = ? \qquad \{a, b, c\} \times \{d, e\} = ?$$

Often these operators are defined to give it properties that are consistent with its usage elsewhere, e.g. 'commutativity': $\mathbf{a} + \mathbf{b} = \mathbf{b} + \mathbf{a}$ for vector addition just as $2 + 4 = 4 + 2$ for numbers.

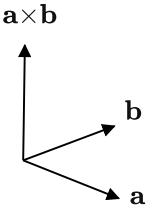
'Multiplying' **two sets** (known again as the cross product) finds each possible combination of members, one from each:

$$\{a, b, c\} \times \{d, e\} = \{\{a, d\}, \{a, e\}, \{b, d\}, \{b, e\}, \{c, d\}, \{c, e\}\}$$

It has the nice property that $n(A \times B) = n(A) \times n(B)$, where $n(A)$ gives the size of the set A. Also, $A \times B = B \times A$.

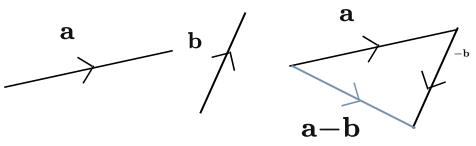
In FM, you will see that 'multiplying' two 3D vectors (known as the **cross product**) gives you a vector **perpendicular to the two**.
Vector multiplication is not commutative, so $\mathbf{a} \times \mathbf{b} \neq \mathbf{b} \times \mathbf{a}$ (however it is 'distributive', so $\mathbf{a} \times (\mathbf{b} + \mathbf{c}) = (\mathbf{a} \times \mathbf{b}) + (\mathbf{a} \times \mathbf{c})$)
The Casio Classwiz can calculate this in Vector mode!

$$\begin{pmatrix} 3 \\ 2 \\ -1 \end{pmatrix} \times \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ -4 \\ 2 \end{pmatrix}$$



E Vector **subtraction** is defined using vector addition and negation:

$$\mathbf{a} - \mathbf{b} = \mathbf{a} + (-\mathbf{b})$$



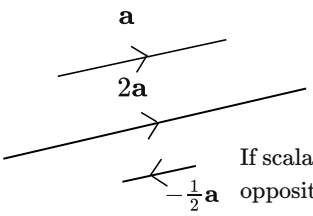
F The zero vector **0** (a bold 0), represents no movement.

$$\vec{PQ} + \vec{QP} = \mathbf{0}$$

In 2D: $\mathbf{0} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

G A **scalar** is a normal number, which can be used to 'scale' a vector.

- The **direction** will be the **same**.
- But the **magnitude** will be **different** (unless the scalar is 1).



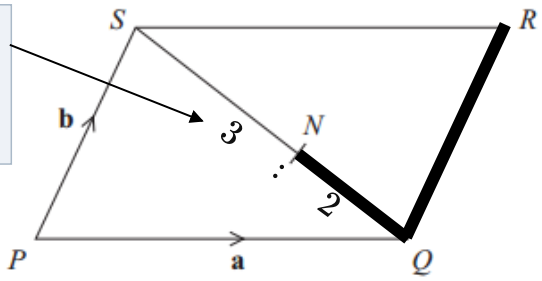
If scalar is negative, opposite direction.

H Any vector parallel to the vector **a** can be written as $\lambda\mathbf{a}$, where λ is a scalar.

The implication is that if we can write one vector **as a multiple of** another, then we can show they are parallel.

"Show $2\mathbf{a} + 4\mathbf{b}$ and $3\mathbf{a} + 6\mathbf{b}$ are parallel".
$3\mathbf{a} + 6\mathbf{b} = \frac{3}{2}(\mathbf{a} + 2\mathbf{b}) \therefore$ parallel

Tip: This ratio wasn't in the original diagram. I like to add the ratio as a visual aid.



$PQRS$ is a parallelogram.

N is the point on SQ such that $SN : NQ = 3 : 2$.

$\overrightarrow{PQ} = \mathbf{a}$ $\overrightarrow{PS} = \mathbf{b}$

a $\vec{SQ} =$

For (b), there's two possible paths to get from N to R: via S or via Q. But which is best?

S Q

b

Tip: start your working with the first line so that your chosen route is clearer.

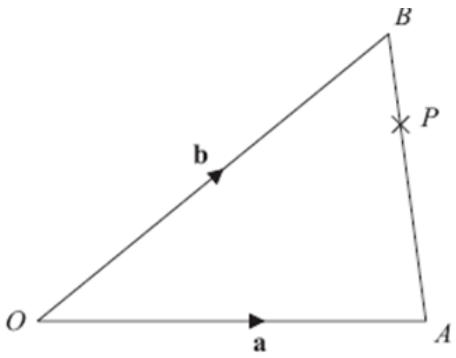


Diagram **NOT** accurately drawn

OAB is a triangle.

$\overrightarrow{OA} = \mathbf{a}$ $\overrightarrow{OB} = \mathbf{b}$

(a) Find $\overrightarrow{AB}$ in terms of **a** and **b**. (1)

P is the point on AB such that $AP : PB = 3 : 1$.

(b) Find $\overrightarrow{OP}$ in terms of **a** and **b**.

Give your answer in its simplest form.

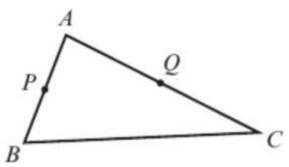
6 $ABCD$ is a trapezium with AB parallel to DC and $DC = 3AB$. M divides DC such that $DM : MC = 2 : 1$. $\overrightarrow{AB} = \mathbf{a}$ and $\overrightarrow{BC} = \mathbf{b}$. Find, in terms of **a** and **b**: **a** $\overrightarrow{AM}$

b $\overrightarrow{BD}$

c $\overrightarrow{MB}$

d $\overrightarrow{DA}$.

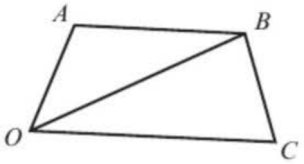
9 In triangle ABC, $\overrightarrow{AB} = \mathbf{a}$ and $\overrightarrow{AC} = \mathbf{b}$.



P is the midpoint of AB and Q is the midpoint of AC.

a Write in terms of **a** and **b**:

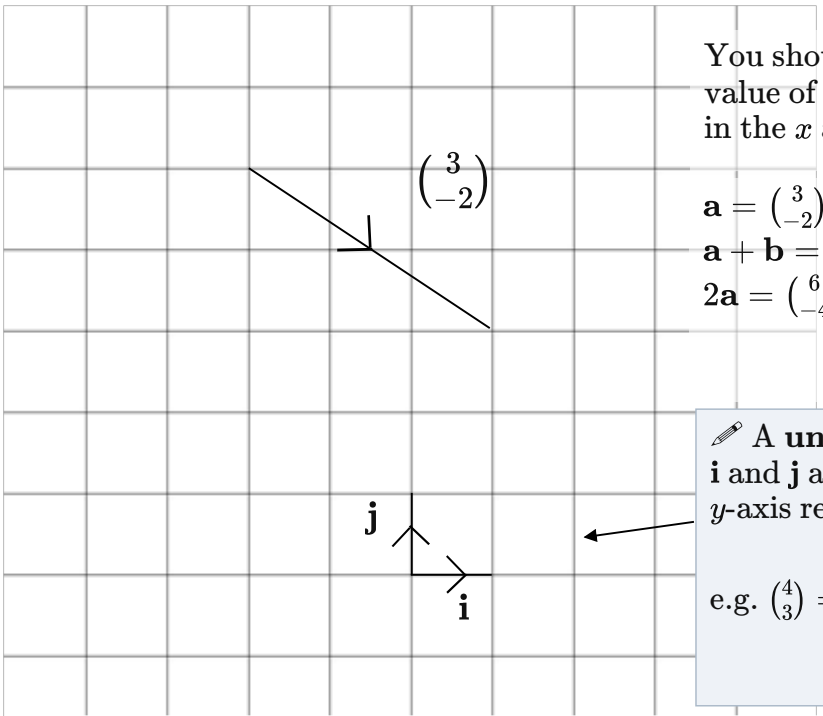
10 $OABC$ is a quadrilateral. $\overrightarrow{OA} = \mathbf{a}$, $\overrightarrow{OC} = 3\mathbf{b}$ and $\overrightarrow{OB} = \mathbf{a} + 2\mathbf{b}$.



a Find, in terms of **a** and **b**:

i $\overrightarrow{AB}$ ii $\overrightarrow{CB}$ iii AB iv OC

11 The vectors $2\mathbf{a} + k\mathbf{b}$ and $5\mathbf{a} + 3\mathbf{b}$ are parallel. Find the value of k.



You should already be familiar that the value of a vector is the **displacement** in the x and y direction (if in 2D).

$\mathbf{a} = \begin{pmatrix} 3 \\ -2 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} 0 \\ -1 \end{pmatrix}$
$\mathbf{a} + \mathbf{b} = \begin{pmatrix} 3 \\ -3 \end{pmatrix}$
$2\mathbf{a} = \begin{pmatrix} 6 \\ -4 \end{pmatrix}$

A **unit vector** is a vector of magnitude 1. $\mathbf{i}$ and $\mathbf{j}$ are unit vectors in the x-axis and y-axis respectively.

$$\mathbf{i} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \mathbf{j} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

e.g. $\begin{pmatrix} 4 \\ 3 \end{pmatrix} = 4\begin{pmatrix} 1 \\ 0 \end{pmatrix} + 3\begin{pmatrix} 0 \\ 1 \end{pmatrix} = 4\mathbf{i} + 3\mathbf{j}$

Notes: This allows us to write any vector algebraically without using vector notation. **Any point in 2D space,** as a vector from the origin, can be obtained using a linear combination of **i** and **j**, e.g. if $P(5, -1)$, $\vec{OP} = 5\mathbf{i} - \mathbf{j}$. For this reason, **i** and **j** are known as **basis vectors** of 2D coordinate space. In fact, any two non-parallel/non-zero vectors can be used as basis vectors, e.g. if $\mathbf{a} = \begin{pmatrix} 5 \\ 2 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 3 \\ 0 \end{pmatrix}$, it's possible to get any vector $\begin{pmatrix} x \\ y \end{pmatrix}$ using a linear combination of these, i.e. we can always find scalars p and q such that $\begin{pmatrix} x \\ y \end{pmatrix} = p\begin{pmatrix} 5 \\ 2 \end{pmatrix} + q\begin{pmatrix} 3 \\ 0 \end{pmatrix}$.

If $\mathbf{a} = 3\mathbf{i}, \quad \mathbf{b} = \mathbf{i} + \mathbf{j}, \quad \mathbf{c} = \mathbf{i} - 2\mathbf{j}$ then:

1) Write **a** in vector form.

2) Find **b**+**2c** in **i**,**j** form.

1

2

$$\mathbf{b} + 2\mathbf{c} = \mathbf{i} + \mathbf{j} + 2(\mathbf{i} - 2\mathbf{j})$$

4 Given that $\mathbf{a} = 2\mathbf{i} + 5\mathbf{j}$ and $\mathbf{b} = 3\mathbf{i} - \mathbf{j}$, find:

a λ if $\mathbf{a} + \lambda\mathbf{b}$ is parallel to the vector $\mathbf{i}$.

b μ if $\mu\mathbf{a} + \mathbf{b}$ is parallel to the vector $\mathbf{j}$.

5 Given that $\mathbf{c} = 3\mathbf{i} + 4\mathbf{j}$ and $\mathbf{d} = \mathbf{i} - 2\mathbf{j}$, find:

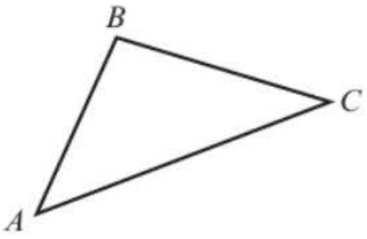
a λ if $\mathbf{c} + \lambda\mathbf{d}$ is parallel to $\mathbf{i} + \mathbf{j}$.

b μ if $\mu\mathbf{c} + \mathbf{d}$ is parallel to $\mathbf{i} + 3\mathbf{j}$.

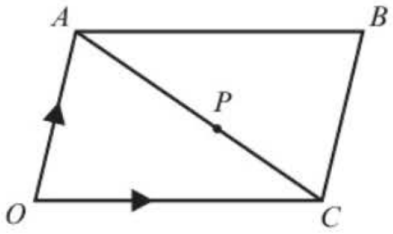
c s if $\mathbf{c} - s\mathbf{d}$ is parallel to $2\mathbf{i} + \mathbf{j}$.

d t if $\mathbf{d} - t\mathbf{c}$ is parallel to $-2\mathbf{i} + 3\mathbf{j}$.

6 In triangle ABC, $\overrightarrow{AB} = 4\mathbf{i} + 3\mathbf{j}$ and $\overrightarrow{AC} = 5\mathbf{i} + 2\mathbf{j}$. Find $\overrightarrow{BC}$.



7 $OABC$ is a parallelogram. P divides AC in the ratio $3 : 2$.
$\overrightarrow{OA} = 2\mathbf{i} + 4\mathbf{j}$, $\overrightarrow{OC} = 7\mathbf{i}$. Find in $\mathbf{i}, \mathbf{j}$ format and column vector format: **a** $\overrightarrow{AC}$
b $\overrightarrow{OP}$
c $\overrightarrow{AP}$.



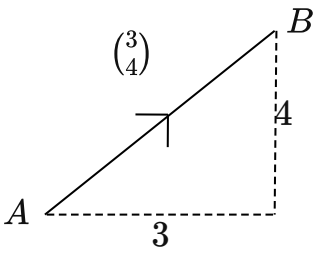
8 $\mathbf{a} = \begin{pmatrix} j \\ 3 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} 10 \\ k \end{pmatrix}$, $\mathbf{c} = \begin{pmatrix} 2 \\ 5 \end{pmatrix}$. Given that $\mathbf{b} - 2\mathbf{a} = \mathbf{c}$, find the values of j and k.

9 $\mathbf{a} = \begin{pmatrix} p \\ -q \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} q \\ p \end{pmatrix}$, $\mathbf{c} = \begin{pmatrix} 7 \\ 4 \end{pmatrix}$. Given that $\mathbf{a} + 2\mathbf{b} = \mathbf{c}$, find the values of p and q.

10 The resultant of the vectors $\mathbf{a} = 3\mathbf{i} - 2\mathbf{j}$ and $\mathbf{b} = p\mathbf{i} - 2p\mathbf{j}$ is parallel to the vector $\mathbf{c} = 2\mathbf{i} - 3\mathbf{j}$. Find:
a the value of p **b** the resultant of vectors **a** and **b**.

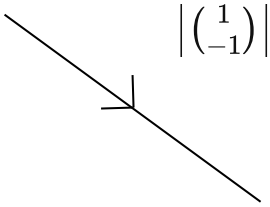
The magnitude $|a|$ of a vector a is its length.

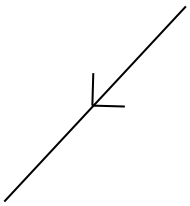
! If $\mathbf{a} = \begin{pmatrix} x \\ y \end{pmatrix}$ $|\mathbf{a}| = \sqrt{x^2 + y^2}$



$\left|\vec{AB}\right| =$

$\left|\begin{pmatrix} 1 \\ -1 \end{pmatrix}\right| =$





$\left|\begin{pmatrix} -5 \\ -12 \end{pmatrix}\right| =$

$\mathbf{a} = \begin{pmatrix} 4 \\ -1 \end{pmatrix}$ $|\mathbf{a}| =$

$\mathbf{b} = \begin{pmatrix} 2 \\ 0 \end{pmatrix}$ $|\mathbf{b}| =$

A unit vector is a vector whose magnitude is 1

There's certain operations on vectors that require the vectors to be 'unit' vectors. We just scale the vector so that its magnitude is now 1.

$\mathbf{a} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$

$|\mathbf{a}| = \sqrt{3^2 + 4^2} = 5$

If $\mathbf{a}$ is a vector, then the unit vector $\hat{\mathbf{a}}$ in the same direction is

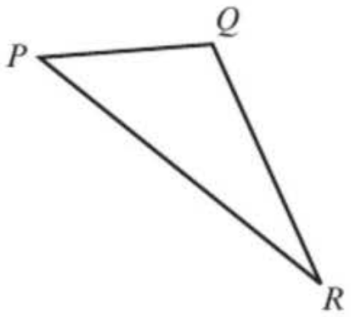
$$\hat{\mathbf{a}} = \frac{\mathbf{a}}{|\mathbf{a}|}$$

Test Your Understanding: Convert the following vectors to unit vectors.

8 Given that $|2\mathbf{i} - k\mathbf{j}| = 2\sqrt{10}$, find the exact value of k.

9 Vector $\mathbf{a} = p\mathbf{i} + q\mathbf{j}$ has magnitude 10 and makes an angle θ with the positive x-axis where $\sin\theta = \frac{3}{5}$. Find the possible values of p and q.

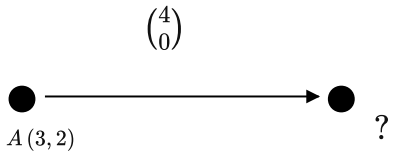
11 In triangle PQR, $\overrightarrow{PQ} = 4\mathbf{i} + \mathbf{j}$, $\overrightarrow{PR} = 6\mathbf{i} - 8\mathbf{j}$.



a Find the size of $\angle QPR$, in degrees, to one decimal place. (5 marks)

b Find the area of triangle PQR. (2 marks)

Suppose we started at a point $(3, 2)$ and translated by the vector $\binom{4}{0}$:



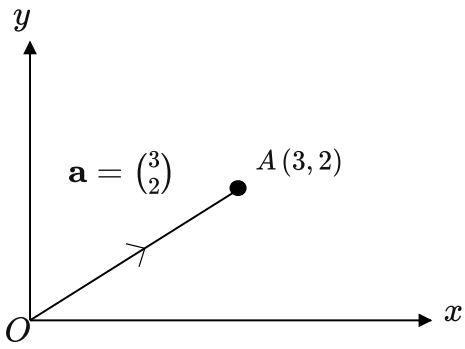
You might think we can do something like:

$$(3, 2) + \binom{4}{0} = (7, 2)$$

But only vectors can be added to other vectors.

If we treated the point **(3,2) as a vector**, then this solves the problem:

$$\binom{3}{2} + \binom{4}{0} = \binom{7}{2}$$



A vector used to represent a position is unsurprisingly known as a **position vector**. A position can be thought of as a translation from the origin, as per above. It enables us to use positions in all sorts of vector (and matrix!) calculations.

The position vector of a point A is the vector $\vec{OA}$, where O is the origin. $\vec{OA}$ is usually written as **a**.

The points A and B have coordinates $(3, 4)$ and $(11, 2)$ respectively.
Find, in terms of i and j:

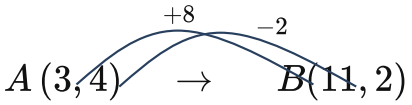
a) The position vector of A
b) The position vector of B
c) The vector $\vec{AB}$

a

b $\vec{OB} =$

c $\vec{AB} =$

You can see this by inspection of the change in x and the change in y:



More formally:

$$\vec{AB} = \vec{OB} - \vec{OA}$$
$$= \binom{11}{2} - \binom{3}{4} = \binom{8}{-2}$$

$\vec{OA} = 5i - 2j$ and $\vec{AB} = 3i + 4j$. Find:
a) The position vector of B.
b) The exact value of $\left|\vec{OB}\right|$ in simplified surd form.

a C

b $\left|\vec{OB}\right| =$

Either a quick sketch will help you see this, or thinking of $\vec{OA}$ as the original position and $\vec{AB}$ as the translation.

4 $OABCDE$ is a regular hexagon. The points A and B have position vectors **a** and **b** respectively, where O is the origin. Find, in terms of **a** and **b**, the position vectors of **a** C
b D
c E.

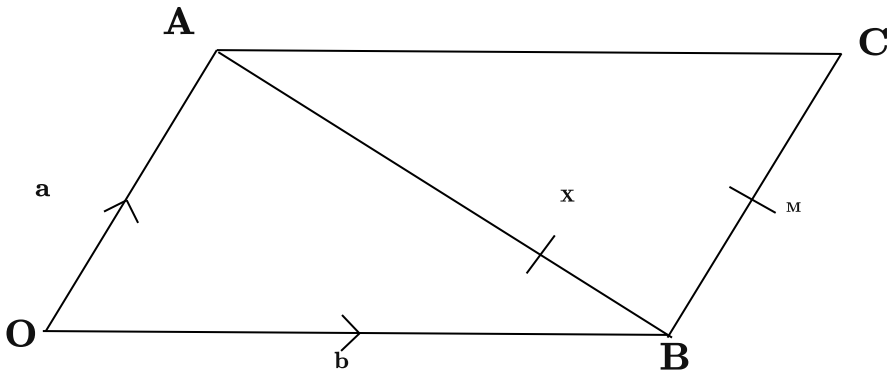
5 The position vectors of 3 vertices of a parallelogram are $\begin{pmatrix}4\\2\end{pmatrix}$, $\begin{pmatrix}3\\5\end{pmatrix}$ and $\begin{pmatrix}8\\6\end{pmatrix}$.
Find the possible position vectors of the fourth vertex.

6 Given that the point A has position vector $4\mathbf{i} - 5\mathbf{j}$ and the point B has position vector $6\mathbf{i} + 3\mathbf{j}$,

a find the vector $\overrightarrow{AB}$. (2 ma

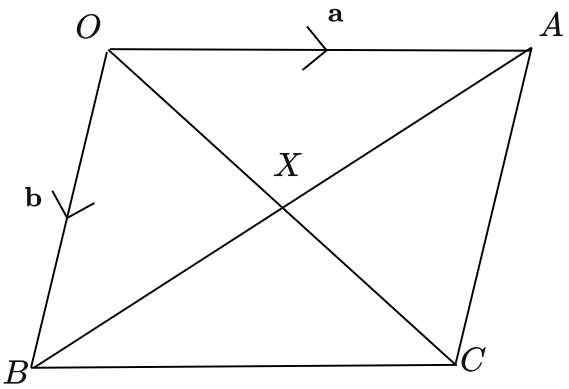
b find $|\overrightarrow{AB}|$ giving your answer as a simplified surd. (2 ma

7 The point A lies on the circle with equation $x^2 + y^2 = 9$. Given that $\overrightarrow{OA} = 2k\mathbf{i} + k\mathbf{j}$, find the exact value of k.

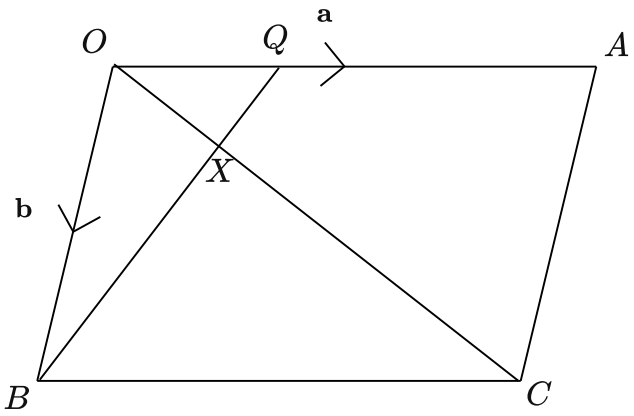


X is a point on AB such that $AX : XB = 3 : 1$. M is the midpoint of BC.
Show that $\vec{XM}$ is parallel to $\vec{OC}$.

Remember when we had **identities** like: $ax^2 + 3x \equiv 2x^2 + bx$
we could **compare coefficients**, so that $a = 2$ and $3 = b$.
We can do the same with (non-parallel) vectors!



$OACB$ is a parallelogram, where $\vec{OA} = a$ and $\vec{OB} = b$. The diagonals OC and AB intersect at a point X. Prove that the diagonals bisect each other.
(Hint: Perhaps find $\vec{OX}$ in two different ways?)



In the above diagram, $\vec{OA} = \mathbf{a}$, $\vec{OB} = \mathbf{b}$ and $\vec{OQ} = \frac{1}{3}\mathbf{a}$. We wish to find the ratio $OX : XC$.

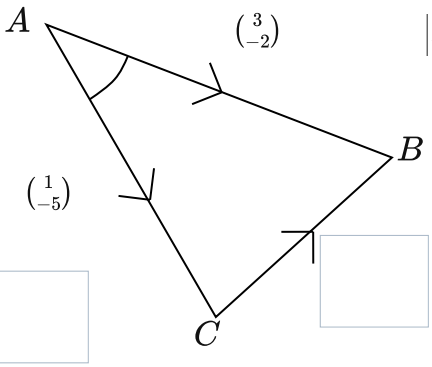
a) If $\vec{OX} = \lambda\vec{OC}$, find an expression for $\vec{OX}$ in terms of **a,b** and λ.
b) If $\vec{BX} = \mu\vec{BQ}$, find an expression for $\vec{OX}$ in terms of **a,b** and μ.
c) By comparing coefficients or otherwise, determine the value of λ, and hence the ratio $OX : XC$.

a

b

c

$\vec{AB} = 3\mathbf{i} - 2\mathbf{j}$ and $\vec{AC} = \mathbf{i} - 5\mathbf{j}$. Determine $\angle BAC$.



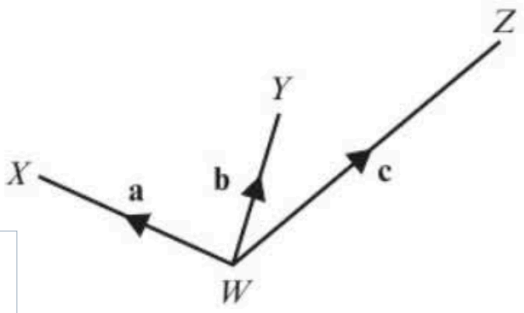
$\left|\vec{AB}\right| = \sqrt{}$

$\left|\vec{AC}\right| = \sqrt{}$

$\left|\vec{CB}\right| = \sqrt{}$

1 In the diagram, $\overrightarrow{WX} = \mathbf{a}$, $\overrightarrow{WY} = \mathbf{b}$ and $\overrightarrow{WZ} = \mathbf{c}$. It is given that $\overrightarrow{XY} = \overrightarrow{YZ}$.

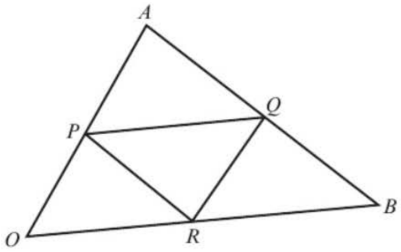
Prove that $\mathbf{a} + \mathbf{c} = 2\mathbf{b}$.



2 OAB is a triangle. P, Q and R are the midpoints of OA, AB and OB respectively. OP and OR are equal to **p** and **r** respectively.

a Find **i** $\overrightarrow{OB}$ **ii** $\overrightarrow{PQ}$.

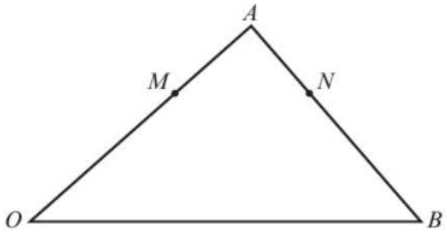
b Hence prove that triangle PAQ is similar to triangle OAB.



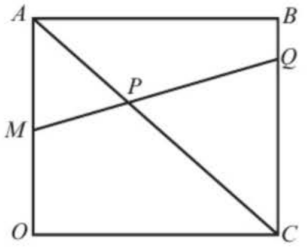
3 OAB is a triangle. $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OB} = \mathbf{b}$. The point M divides OA in the ratio $2 : 1$. MN is parallel to OB.

a Express the vector $\overrightarrow{ON}$ in terms of **a** and **b**.

b Show that $AN : NB = 1 : 2$.

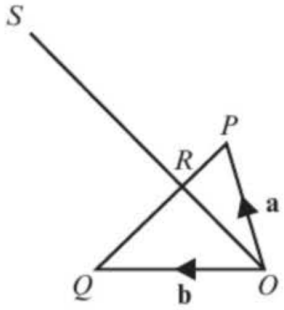


4 $OABC$ is a square. M is the midpoint of OA, and Q divides BC in the ratio $1 : 3$. AC and MQ meet at P.



a If $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OC} = \mathbf{c}$, express $\overrightarrow{OP}$ in terms of $\mathbf{a}$ and $\mathbf{c}$.
b Show that P divides AC in the ratio $2 : 3$.

6 OPQ is a triangle.



$2\overrightarrow{PR} = \overrightarrow{RQ}$ and $3\overrightarrow{OR} = \overrightarrow{OS}$ $\overrightarrow{OP} = \mathbf{a}$ and $\overrightarrow{OQ} = \mathbf{b}$.

a Show that $\overrightarrow{OS} = 2\mathbf{a} + \mathbf{b}$.

b Point T is added to the diagram such that $\overrightarrow{OT} = -\mathbf{b}$.

Prove that points T, P and S lie on a straight line.

1 *[STEP 2010 Q7]*

Relative to a fixed origin O, the points A and B have position vectors $\mathbf{a}$ and $\mathbf{b}$, respectively. (The points O, A and B are not collinear.) The point C has position vector $\mathbf{c}$ given by

$$\mathbf{c} = \alpha\mathbf{a} + \beta\mathbf{b},$$

where α and β are positive constants with $\alpha + \beta < 1$. The lines OA and BC meet at the point P with position vector $\mathbf{p}$ and the lines OB and AC meet at the point Q with position vector $\mathbf{q}$. Show that

$$\mathbf{p} = \tfrac{\alpha a}{1-\beta}$$

and write down q in terms of α, β and b.

Show further that the point R with position vector r given by

$$\mathbf{r} = \tfrac{\alpha\mathbf{a}+\beta\mathbf{b}}{\alpha+\beta},$$

lies on the lines OC and AB.

The lines OB and PR intersect at the point S. Prove that $\frac{OQ}{BQ} = \frac{OS}{BS}$.

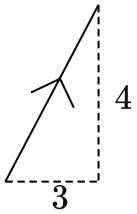
Click here for the solution:
http://www.mathshelper.co.uk/STEP%20Solutions/STEP%20Solutions.pdf
(STEP 2010 Q7)

In Mechanics, you will see certain things can be represented as a simple number (without direction), or as a vector (with direction):

Remember a 'scalar' just means a normal number (in the context of vectors). It can be obtained using the **magnitude** of the vector.

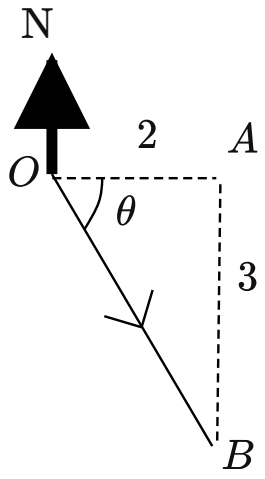
Vector Quantity	**Equivalent Scalar Quantity**
Velocity e.g. $\begin{pmatrix}3\\4\end{pmatrix} km/h$	
Displacement e.g. $\begin{pmatrix}-5\\12\end{pmatrix} km$	

This means the position vector of the object changes by $\begin{pmatrix}3\\4\end{pmatrix}$ each hour.



[Textbook] A girl walks 2 km due east from a fixed point O to A, and then 3 km due south from A to B. Find

a) the total distance travelled
b) the position vector of B relative to O
c) $\left|\vec{OB}\right|$
d) The bearing of B from O.



a

b

$$\left|\vec{OB}\right| = \sqrt{2^2 + 3^2} = \sqrt{13} = 3.61km$$

c

d

[Textbook] In an orienteering exercise, a cadet leaves the starting point O and walks 15 km on a bearing of 120° to reach A, the first checkpoint. From A he walks 9 km on a bearing of 240° to the second checkpoint, at B. From B he returns directly to O.
Find:

a) the position vector of A relative to O
b) $\left|\vec{OB}\right|$
c) the bearing of B from O
d) the position vector of B relative O.

I have no specific advice to offer except:

1. Draw a BIG diagram.
2. Remember bearings are measured clockwise from North.
3. Don't forget units (on vectors!)

a

b

c

Further Example 37

[Textbook] In an orienteering exercise, a cadet leaves the starting point O and walks 15 km on a bearing of 120° to reach A, the first checkpoint. From A he walks 9 km on a bearing of 240° to the second checkpoint, at B. From B he returns directly to O.
Find:

a) the position vector of A relative to O
b) $\left|\vec{OB}\right|$
c) the bearing of B from O
d) the position vector of B relative O.

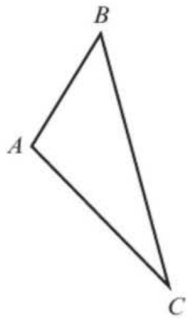
d

Exercise 11F - Year 1 Pages 250-251 38

5 A particle P of mass $m = 0.3$ kg moves under the action of a single constant force $\mathbf{F}$ newtons. The acceleration of P is $\mathbf{a} = (5\mathbf{i} + 7\mathbf{j})$ m s^{-2}.
a Find the angle between the acceleration and $\mathbf{i}$. (2 ma
Force, mass and acceleration are related by the formula $\mathbf{F} = m\mathbf{a}$.
b Find the magnitude of $\mathbf{F}$. (3 ma

6 Two forces, $\mathbf{F}_1$ and $\mathbf{F}_2$, are given by the vectors $\mathbf{F}_1 = (3\mathbf{i} - 4\mathbf{j})$ N and $\mathbf{F}_2 = (p\mathbf{i} + q\mathbf{j})$ N. The resultant force, $\mathbf{R} = \mathbf{F}_1 + \mathbf{F}_2$ acts in a direction which is parallel to the vector $(2\mathbf{i} - \mathbf{j})$.
a Find the angle between $\mathbf{R}$ and the vector $\mathbf{i}$.
b Show that $p + 2q = 5$.
c Given that $p = 1$, find the magnitude of $\mathbf{R}$.

7 The diagram shows a sketch of a field in the shape of a triangle ABC. Given $\overrightarrow{AB} = 30\mathbf{i} + 40\mathbf{j}$ metres and $\overrightarrow{AC} = 40\mathbf{i} - 60\mathbf{j}$ metres,



a find $\overrightarrow{BC}$ (2 ma
b find the size of $\angle BAC$, in degrees, to one decimal place (4 ma
c find the area of the field in square metres. (3 ma

8 A boat has a position vector of $(2\mathbf{i} + \mathbf{j})$ km and a buoy has a position vector of $(6\mathbf{i} - 4\mathbf{j})$ km, relative to a fixed origin O.
a Find the distance of the boat from the buoy.
b Find the bearing of the boat from the buoy.
The boat travels with constant velocity $(8\mathbf{i} - 10\mathbf{j})$ km/h.
c Verify that the boat is travelling directly towards the buoy.
d Find the speed of the boat.
e Work out how long it will take the boat to reach the buoy.
Problem-solving Draw a sketch showing t
initial positions of the bo
the buoy and the origin.